

Source folder: /Users/jonat/Documents/GitHub/master_thesis
Files in this part: 16

File: results_old_broken/dp_inf_alpha_0.5_dim_16_clients_100_seed_123_summary.csv

```
Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients,Total_Samples
1,0.47193825115015936,0.0,0.08,0.008,0.0010183259698601948,0.4507420846236263,0.450919524819384,0.549246813
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File: results_old_broken/dp_inf_alpha_0.5_dim_16_clients_100_seed_42.json

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    "use_dp": false,
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    "dp_clip_norm": 1.0,
    "num_rounds": 10,
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    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 42
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        "accuracy": 0.5492468134414832,
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        "Precision@10": 0.008,
        "Recall@10": 0.0006694849317098337,
        "num_users_evaluated": 50
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  "loss": 0.6540295489789542,
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  "loss": 0.6071505906920313,
  "samples": 92,
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  "loss": 0.4479467417210566,
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      "samples": 76,
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    "mse": 0.45077402767134983,
    "mae": 0.4509407244579541,
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"Recall@10": 0.0006694849317098337,
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    "samples": 92,
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      "loss": 0.5237976354784554,
      "samples": 85,
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    "mae": 0.45094072114274786,
    "accuracy": 0.5492468134414832,
    "samples": 1726,
    "NDCG@10": 0.0,
    "Hit@10": 0.08,
    "Precision@10": 0.008,
    "Recall@10": 0.0006694849317098337,
    "num_users_evaluated": 50
  },
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    "total_samples": 6706
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      "samples": 53,
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    {
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      "loss": 0.6071522794897369,
      "samples": 92,
      "epochs": 1
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      "samples": 56,
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  "loss": 0.555621972030704,
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  "samples": 42,
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  "samples": 47,
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{
  "client_id": "client_92",
  "loss": 0.37503012144680753,
  "samples": 64,
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{
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  "loss": 0.6385545220027224,
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        "samples": 37,
        "epochs": 1
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    {
        "client_id": "client_95",
        "loss": 0.7166133218176218,
        "samples": 46,
        "epochs": 1
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    {
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    {
        "client_id": "client_98",
        "loss": 0.5183398783942044,
        "samples": 85,
        "epochs": 1
    },
    {
        "client_id": "client_99",
        "loss": 0.4096737429860259,
        "samples": 76,
        "epochs": 1
    }
    ],
    "timestamp": "2025-12-02T12:52:23.006936"
}
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    "mae": 0.45094072114274786,
    "accuracy": 0.5492468134414832,
    "samples": 1726,
    "NDCG@10": 0.0,
    "Hit@10": 0.08,
    "Precision@10": 0.008,
    "Recall@10": 0.0006694849317098337,
    "num_users_evaluated": 50
},
"client_metrics": [],
"mobile_metrics": []
}

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File: results_old_broken/dp_inf_alpha_0.5_dim_16_clients_100_seed_42_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients,Total_Samples
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134414832,97,6706
3,0.46844184908375225,0.0,0.08,0.008,0.0006694849317098337,0.4507740199358687,0.4509407222478166,0.54924681
34414832,97,6706
4,0.47262820705578074,0.0,0.08,0.008,0.0006694849317098337,0.4507740122003876,0.4509407277316024,0.5492468
134414832,97,6706
5,0.4699316094649674,0.0,0.08,0.008,0.0006694849317098337,0.4507740023238358,0.4509407189326104,0.549246813
4414832,97,6706
6,0.47724114365328313,0.0,0.08,0.008,0.0006694849317098337,0.4507740033598377,0.45094072114274786,0.5492468
134414832,97,6706
7,0.4760963835744217,0.0,0.08,0.008,0.0006694849317098337,0.45077399783449407,0.4509407156174042,0.54924681

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34414832,97,6706
8,0.47854143835143315,0.0,0.08,0.008,0.0006694849317098337,0.45077400011369834,0.45094071451233547,0.549246
8134414832,97,6706
9,0.4687992223147701,0.0,0.08,0.008,0.0006694849317098337,0.45077399893956277,0.450940712302198,0.549246813
4414832,97,6706
10,0.4734033131578739,0.0,0.08,0.008,0.0006694849317098337,0.45077399893956277,0.45094072114274786,0.549246
8134414832,97,6706
FINAL,,0.0,0.08,0.008,0.0006694849317098337,0.45077399893956277,0.45094072114274786,0.5492468134414832,,
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File: results_old_broken/dp_inf_alpha_0.5_dim_16_clients_100_seed_456.json

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    "num_clients": 100,
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    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 1.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 1,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 456
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  "rounds": [
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      "test_metrics": {
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        "mae": 0.4509111141412587,
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        "Hit@10": 0.02,
        "Precision@10": 0.004,
        "Recall@10": 0.0009523809523809524,
        "num_users_evaluated": 50
      },
      "aggregation": {
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        "round": 200,
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        "total_samples": 6706
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          "loss": 0.546859154856975,
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        {
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          "loss": 0.6539951571161147,
          "samples": 53,
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        },
        {
          "client_id": "client_3",
          "loss": 0.6026206292649813,
          "samples": 92,
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          "client_id": "client_4",
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          "samples": 56,
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  "loss": 0.47061808117903864,
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{
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  "loss": 0.4530668377518594,
  "samples": 64,
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  "samples": 75,
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  "loss": 0.40022822149522125,
  "samples": 122,
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  "loss": 0.3203763489663226,
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},
{
  "client_id": "client_10",
  "loss": 0.41671988696838785,
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},
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  "loss": 0.40626565170268436,
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{
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  "loss": 0.44352023391373807,
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},
{
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  "loss": 0.6124786056373756,
  "samples": 42,
  "epochs": 1
},
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        "loss": 0.42713845439366277,
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    "Hit@10": 0.02,
    "Precision@10": 0.004,
    "Recall@10": 0.0009523809523809524,
    "num_users_evaluated": 50
},
"client_metrics": [],
"mobile_metrics": []
}

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File: results_old_broken/dp_inf_alpha_0.5_dim_16_clients_100_seed_456_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients,Total_Samples
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34414832,97,6706
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File: results_old_broken/dp_inf_alpha_1.0_dim_16_clients_100_seed_123.json

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        "num_clients": 100,
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"batch_size": 32,
"seed": 123
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"rounds": [
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      "Hit@10": 0.08,
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      "Recall@10": 0.0011231016970147402,
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        "loss": 0.43891360271727087,
        "samples": 54,
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      {
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        "loss": 0.30331257802763306,
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    "Recall@10": 0.0011231016970147402,
    "num_users_evaluated": 50
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"client_metrics": [],
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File: results_old_broken/dp_inf_alpha_1.0_dim_16_clients_100_seed_123_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients>Total_Samples
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34414832,99,6804
2,0.4632825547538466,0.0,0.08,0.008,0.0011231016970147402,0.4507457838412094,0.45089501549995925,0.54924681
34414832,99,6804
3,0.4656622521994105,0.0,0.08,0.008,0.0011231016970147402,0.45074578163107193,0.4508950199202342,0.54924681

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34414832,99,6804
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File: results_old_broken/dp_inf_alpha_1.0_dim_16_clients_100_seed_42.json

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    "use_dp": false,
    "dp_sigma": 1.0,
    "dp_clip_norm": 1.0,
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    "batch_size": 32,
    "seed": 42
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        "num_users_evaluated": 50
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  "accuracy": 0.5492468134414832,
  "samples": 1726,
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  "Hit@10": 0.06,
  "Precision@10": 0.006000000000000001,
  "Recall@10": 0.0030572862582208377,
  "num_users_evaluated": 50
},
"client_metrics": [],
"mobile_metrics": []
}
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File: results_old_broken/dp_inf_alpha_1.0_dim_16_clients_100_seed_42_summary.csv

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Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients>Total_Samples
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2,0.4549620682659051,0.0,0.06,0.006000000000000001,0.0030572862582208377,0.45074820642813884,0.450946039838
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3,0.460683318768338,0.0,0.06,0.006000000000000001,0.0030572862582208377,0.4507482140945532,0.45094604646897
124,0.5492468134414832,99,6804
4,0.46340030014016154,0.0,0.06,0.006000000000000001,0.0030572862582208377,0.45074820974334506,0.45094605088
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6,0.4611339805770383,0.0,0.06,0.006000000000000001,0.0030572862582208377,0.4507482152686887,0.4509460431537
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134414832,,
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File: results_old_broken/dp_inf_alpha_1.0_dim_16_clients_100_seed_456.json

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  "timestamp": "2025-12-02T19:49:15.867683",
  "config": {
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    "embedding_dim": 16,
    "num_clients": 100,
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    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 1.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 1,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 456
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  "rounds": [
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        "accuracy": 0.5492468134414832,

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"Hit@10": 0.06,
"Precision@10": 0.006000000000000001,
"Recall@10": 0.004167341430499325,
"num_users_evaluated": 50
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"aggregation": {
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  "round": 230,
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    "loss": 0.405388796303761,
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    "loss": 0.4530602812442124,
    "samples": 54,
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    "loss": 0.5698650467246973,
    "samples": 81,
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    "loss": 0.5668549634963639,
    "samples": 71,
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    "loss": 0.45995960483426923,
    "samples": 60,
    "epochs": 1
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  {
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    "loss": 0.45832246165722734,
    "samples": 84,
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    "loss": 0.5540614636060938,
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  {
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{
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{
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  "samples": 87,
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{
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  "loss": 0.34376110829846956,
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  "client_id": "client_78",
  "loss": 0.32497554119060734,
  "samples": 84,
  "epochs": 1
},
{
  "client_id": "client_79",
  "loss": 0.5128010182567403,
  "samples": 49,
  "epochs": 1
},
{
  "client_id": "client_80",
  "loss": 0.1935657413307278,
  "samples": 31,
  "epochs": 1
},
{
  "client_id": "client_81",
  "loss": 0.6669024489568385,
  "samples": 66,
  "epochs": 1
},
},
```

```
{
  "client_id": "client_82",
  "loss": 0.4026967583868802,
  "samples": 81,
  "epochs": 1
},
{
  "client_id": "client_83",
  "loss": 0.3751232983611642,
  "samples": 64,
  "epochs": 1
},
{
  "client_id": "client_84",
  "loss": 0.4584355614329887,
  "samples": 66,
  "epochs": 1
},
{
  "client_id": "client_85",
  "loss": 0.4342888040114031,
  "samples": 87,
  "epochs": 1
},
{
  "client_id": "client_86",
  "loss": 0.34376098908923963,
  "samples": 64,
  "epochs": 1
},
{
  "client_id": "client_87",
  "loss": 0.4964420101369966,
  "samples": 58,
  "epochs": 1
},
{
  "client_id": "client_88",
  "loss": 0.5283011026497342,
  "samples": 78,
  "epochs": 1
},
{
  "client_id": "client_89",
  "loss": 0.5105034894778865,
  "samples": 80,
  "epochs": 1
},
{
  "client_id": "client_90",
  "loss": 0.5312485422828334,
  "samples": 48,
  "epochs": 1
},
{
  "client_id": "client_92",
  "loss": 0.42011440697610075,
  "samples": 67,
  "epochs": 1
},
{
  "client_id": "client_93",
  "loss": 0.4934210247328546,
  "samples": 63,
  "epochs": 1
},
{
  "client_id": "client_94",
  "loss": 0.5551451351109193,
  "samples": 51,
  "epochs": 1
},
{
  "client_id": "client_95",
```

```
        "loss": 0.5624647308411513,
        "samples": 54,
        "epochs": 1
    },
    {
        "client_id": "client_96",
        "loss": 0.4652297833510127,
        "samples": 79,
        "epochs": 1
    },
    {
        "client_id": "client_97",
        "loss": 0.5186747848415703,
        "samples": 57,
        "epochs": 1
    },
    {
        "client_id": "client_98",
        "loss": 0.40055637478499384,
        "samples": 103,
        "epochs": 1
    },
    {
        "client_id": "client_99",
        "loss": 0.50540842177432,
        "samples": 61,
        "epochs": 1
    }
],
"timestamp": "2025-12-02T21:33:22.124053"
}
],
"final_metrics": {
    "mse": 0.4507318038929379,
    "mae": 0.4509109461708113,
    "accuracy": 0.5492468134414832,
    "samples": 1726,
    "NDCG@10": 0.0,
    "Hit@10": 0.06,
    "Precision@10": 0.006000000000000001,
    "Recall@10": 0.004167341430499325,
    "num_users_evaluated": 50
},
"client_metrics": [],
"mobile_metrics": []
}
```

File: results_old_broken/dp_inf_alpha_1.0_dim_16_clients_100_seed_456_summary.csv

```
Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE,Accuracy,Num_Clients>Total_Samples
1,0.46483067124610283,0.0,0.06,0.006000000000000001,0.004167341430499325,0.45073179512145484,0.450910961641
7736,0.5492468134414832,99,6804
2,0.470360631373535,0.0,0.06,0.006000000000000001,0.004167341430499325,0.4507317918753154,0.450910959431636
1,0.5492468134414832,99,6804
3,0.4659224251276084,0.0,0.06,0.006000000000000001,0.004167341430499325,0.45073179512145484,0.4509109572214
9865,0.5492468134414832,99,6804
4,0.4629648949907534,0.0,0.06,0.006000000000000001,0.004167341430499325,0.45073179284225057,0.4509109561164
299,0.5492468134414832,99,6804
5,0.4623915318127114,0.0,0.06,0.006000000000000001,0.004167341430499325,0.4507317940163861,0.45091096053670
48,0.5492468134414832,99,6804
6,0.46298127242127607,0.0,0.06,0.006000000000000001,0.004167341430499325,0.4507317918062486,0.4509109572214
9865,0.5492468134414832,99,6804
7,0.4571008709020071,0.0,0.06,0.006000000000000001,0.004167341430499325,0.45073179843666106,0.4509109561164
299,0.5492468134414832,99,6804
8,0.4661526531692389,0.0,0.06,0.006000000000000001,0.004167341430499325,0.4507317940163861,0.45091095943163
61,0.5492468134414832,99,6804
9,0.4586607793552642,0.0,0.06,0.006000000000000001,0.004167341430499325,0.4507317918062486,0.45091095501136
12,0.5492468134414832,99,6804
10,0.4625127050566011,0.0,0.06,0.006000000000000001,0.004167341430499325,0.4507318038929379,0.4509109461708
113,0.5492468134414832,99,6804
FINAL,,0.0,0.06,0.006000000000000001,0.004167341430499325,0.4507318038929379,0.4509109461708113,0.549246813
4414832,,
```

File: scripts/_init_.py

```
# Scripts package for federated learning experiments
```

File: scripts/attack_evaluation.py

```
"""
Privacy Attack Evaluation Framework

Implements simplified versions of:
1. Membership Inference Attack (MIA)
2. Model Inversion Attack

Note: This is a simplified implementation. For production use, consider using
AIJack or other established attack libraries.
"""

import torch
import torch.nn as nn
import numpy as np
from typing import List, Dict, Tuple
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import roc_auc_score, accuracy_score
import logging

logger = logging.getLogger(__name__)

class MembershipInferenceAttack:
    """
    Simplified Membership Inference Attack (MIA).

    Uses shadow models to train an attack classifier that predicts
    whether a sample was in the training set.
    """

    def __init__(self, num_shadow_models: int = 5):
        """
        Initialize MIA attack.

        Args:
            num_shadow_models: Number of shadow models to train
        """
        self.num_shadow_models = num_shadow_models
        self.attack_classifier = None

    def extract_features(self, model: nn.Module, sample: Tuple[int, int, float]) -> np.ndarray:
        """
        Extract features from model for a given sample.

        Args:
            model: Trained model
            sample: (user_id, item_id, rating) tuple

        Returns:
            Feature vector
        """
        model.eval()
        user_id, item_id, rating = sample

        with torch.no_grad():
            # Get prediction
            pred = model(torch.LongTensor([user_id]), torch.LongTensor([item_id]))
            pred_value = pred.item()

            # Get embedding vectors
            user_emb = model.user_embedding(torch.LongTensor([user_id])).squeeze().numpy()
            item_emb = model.item_embedding(torch.LongTensor([item_id])).squeeze().numpy()

            # Feature vector: prediction, prediction error, embedding norms, dot product
            features = np.concatenate([
                [pred_value],
                [abs(pred_value - rating)],
```

```

        [np.linalg.norm(user_emb)],
        [np.linalg.norm(item_emb)],
        [np.dot(user_emb, item_emb)],
        user_emb,
        item_emb
    ])

    return features

def train_attack_classifier(self,
                            shadow_models: List[nn.Module],
                            member_data: List[Tuple],
                            non_member_data: List[Tuple]) -> float:
    """
    Train attack classifier using shadow models.

    Args:
        shadow_models: List of shadow models
        member_data: Data that was used to train shadow models
        non_member_data: Data that was not used to train shadow models

    Returns:
        Training accuracy
    """
    # Extract features for members and non-members
    member_features = []
    non_member_features = []

    for model in shadow_models:
        # Sample some member and non-member data
        member_samples = np.random.choice(len(member_data), min(50, len(member_data)), replace=False)
        non_member_samples = np.random.choice(len(non_member_data), min(50, len(non_member_data)), repl
ace=False)

        for idx in member_samples:
            features = self.extract_features(model, member_data[idx])
            member_features.append(features)

        for idx in non_member_samples:
            features = self.extract_features(model, non_member_data[idx])
            non_member_features.append(features)

    if len(member_features) == 0 or len(non_member_features) == 0:
        logger.warning("Not enough data for attack classifier training")
        return 0.0

    # Prepare training data
    X = np.array(member_features + non_member_features)
    y = np.array([1] * len(member_features) + [0] * len(non_member_features))

    # Train classifier
    self.attack_classifier = RandomForestClassifier(n_estimators=50, random_state=42)
    self.attack_classifier.fit(X, y)

    # Training accuracy
    train_pred = self.attack_classifier.predict(X)
    train_acc = accuracy_score(y, train_pred)

    return train_acc

def evaluate(self,
            target_model: nn.Module,
            member_data: List[Tuple],
            non_member_data: List[Tuple]) -> Dict[str, float]:
    """
    Evaluate attack on target model.

    Args:
        target_model: Target model to attack
        member_data: Data that was in training set
        non_member_data: Data that was not in training set

    Returns:
        Dictionary with attack metrics (AUC, accuracy)
    """

```

```

"""
if self.attack_classifier is None:
    logger.error("Attack classifier not trained. Call train_attack_classifier first.")
    return {"auc": 0.0, "accuracy": 0.0}

# Extract features
member_features = []
non_member_features = []

# Sample data for evaluation
member_samples = np.random.choice(len(member_data), min(100, len(member_data)), replace=False)
non_member_samples = np.random.choice(len(non_member_data), min(100, len(non_member_data)), replace
=False)

for idx in member_samples:
    features = self.extract_features(target_model, member_data[idx])
    member_features.append(features)

for idx in non_member_samples:
    features = self.extract_features(target_model, non_member_data[idx])
    non_member_features.append(features)

if len(member_features) == 0 or len(non_member_features) == 0:
    return {"auc": 0.0, "accuracy": 0.0}

# Predictions
X_test = np.array(member_features + non_member_features)
y_test = np.array([1] * len(member_features) + [0] * len(non_member_features))

# Get probabilities for AUC
y_pred_proba = self.attack_classifier.predict_proba(X_test)[: , 1]
y_pred = self.attack_classifier.predict(X_test)

# Metrics
try:
    auc = roc_auc_score(y_test, y_pred_proba)
except:
    auc = 0.0

accuracy = accuracy_score(y_test, y_pred)

return {
    "auc": auc,
    "accuracy": accuracy,
    "true_positive_rate": np.mean(y_pred[y_test == 1] == 1),
    "false_positive_rate": np.mean(y_pred[y_test == 0] == 1)
}

class ModelInversionAttack:
    """
    Simplified Model Inversion Attack.

    Attempts to reconstruct training data from model parameters.
    """

    def __init__(self, num_items: int, top_k: int = 10):
        """
        Initialize model inversion attack.

        Args:
            num_items: Number of items in the dataset
            top_k: Top-K items to consider for reconstruction
        """
        self.num_items = num_items
        self.top_k = top_k

    def reconstruct_user_preferences(self,
                                     model: nn.Module,
                                     user_id: int,
                                     known_items: List[int],
                                     known_ratings: List[float]) -> List[int]:
        """
        Attempt to reconstruct user's top-K preferred items.

```

```

    Args:
        model: Trained model
        user_id: User ID
        known_items: Items we know the user interacted with
        known_ratings: Ratings for known items

    Returns:
        List of top-K predicted item IDs
    """
    model.eval()

    # Get user embedding
    with torch.no_grad():
        user_emb = model.user_embedding(torch.LongTensor([user_id])).squeeze()

        # Predict ratings for all items
        item_ids = torch.arange(self.num_items)
        predictions = []

        for item_id in item_ids:
            pred = model(torch.LongTensor([user_id]), item_id.unsqueeze(0))
            predictions.append(pred.item())

    # Get top-K items
    top_k_indices = np.argsort(predictions)[-self.top_k:][::-1]
    top_k_items = top_k_indices.tolist()

    return top_k_items

def evaluate(self,
            model: nn.Module,
            test_data: List[Tuple],
            ground_truth: Dict[int, List[int]]) -> Dict[str, float]:
    """
    Evaluate inversion attack.

    Args:
        model: Trained model
        test_data: Test data (user_id, item_id, rating)
        ground_truth: Ground truth top-K items per user

    Returns:
        Dictionary with attack metrics (top-K accuracy)
    """
    # Group test data by user
    user_data = {}
    for user_id, item_id, rating in test_data:
        if user_id not in user_data:
            user_data[user_id] = []
        user_data[user_id].append((item_id, rating))

    # Reconstruct for each user
    correct_predictions = 0
    total_users = 0

    for user_id in list(user_data.keys())[:min(50, len(user_data))]: # Limit to 50 users
        if user_id not in ground_truth:
            continue

        user_items = [item for item, _ in user_data[user_id]]
        user_ratings = [rating for _, rating in user_data[user_id]]

        # Reconstruct top-K
        reconstructed = self.reconstruct_user_preferences(
            model, user_id, user_items, user_ratings
        )

        # Check accuracy
        true_top_k = ground_truth.get(user_id, [])
        if len(true_top_k) > 0:
            overlap = len(set(reconstructed) & set(true_top_k))
            if overlap >= self.top_k * 0.2: # At least 20% overlap
                correct_predictions += 1

```

```

        total_users += 1

    top_k_accuracy = correct_predictions / total_users if total_users > 0 else 0.0

    return {
        "top_k_accuracy": top_k_accuracy,
        "top_k": self.top_k,
        "num_users_evaluated": total_users
    }

def run_attack_evaluation(target_model: nn.Module,
                        train_data: List[Tuple],
                        test_data: List[Tuple],
                        shadow_models: List[nn.Module] = None) -> Dict:
    """
    Run both MIA and model inversion attacks.

    Args:
        target_model: Target model to attack
        train_data: Training data (members)
        test_data: Test data (non-members)
        shadow_models: Optional shadow models for MIA

    Returns:
        Dictionary with attack results
    """
    results = {}

    # Membership Inference Attack
    print("Running Membership Inference Attack...")
    mia = MembershipInferenceAttack(num_shadow_models=5)

    if shadow_models is not None and len(shadow_models) > 0:
        # Train attack classifier
        train_acc = mia.train_attack_classifier(shadow_models, train_data, test_data)
        print(f" MIA training accuracy: {train_acc:.4f}")

        # Evaluate
        mia_results = mia.evaluate(target_model, train_data, test_data)
        results["mia"] = mia_results
        print(f" MIA AUC: {mia_results['auc']:.4f}")
        print(f" MIA Accuracy: {mia_results['accuracy']:.4f}")
    else:
        results["mia"] = {"auc": 0.0, "accuracy": 0.0, "note": "No shadow models provided"}

    # Model Inversion Attack
    print("Running Model Inversion Attack...")
    inversion = ModelInversionAttack(num_items=1000, top_k=10) # Adjust num_items as needed

    # Create ground truth (top-K items per user from training data)
    user_items = {}
    for user_id, item_id, rating in train_data:
        if user_id not in user_items:
            user_items[user_id] = []
        user_items[user_id].append((item_id, rating))

    ground_truth = {}
    for user_id, items in user_items.items():
        # Sort by rating and get top-K
        sorted_items = sorted(items, key=lambda x: x[1], reverse=True)
        top_k = [item_id for item_id, _ in sorted_items[:10]]
        ground_truth[user_id] = top_k

    inversion_results = inversion.evaluate(target_model, test_data, ground_truth)
    results["inversion"] = inversion_results
    print(f" Inversion Top-K Accuracy: {inversion_results['top_k_accuracy']:.4f}")

    return results

```


File: scripts/metrics_collector.py

```
"""
Metrics Collector for Federated Learning Experiments
Collects and saves all experimental metrics to JSON/CSV for thesis analysis.
"""

import json
import csv
import os
from datetime import datetime
from typing import Dict, List, Any, Optional
from pathlib import Path
import logging

logger = logging.getLogger(__name__)

class MetricsCollector:
    """Collects and stores experimental metrics"""

    def __init__(self, experiment_id: str, results_dir: str = "results"):
        """
        Initialize metrics collector.

        Args:
            experiment_id: Unique identifier for this experiment
            results_dir: Directory to save results
        """
        self.experiment_id = experiment_id
        self.results_dir = Path(results_dir)
        self.results_dir.mkdir(parents=True, exist_ok=True)

        # Store all collected metrics
        self.experiment_data = {
            "experiment_id": experiment_id,
            "timestamp": datetime.now().isoformat(),
            "config": {},
            "rounds": [],
            "final_metrics": {},
            "client_metrics": [],
            "mobile_metrics": []
        }

        logger.info(f"Initialized metrics collector for experiment: {experiment_id}")

    def set_config(self, config: Dict[str, Any]):
        """
        Set experiment configuration.

        Args:
            config: Dictionary with experiment parameters
        """
        self.experiment_data["config"] = config
        logger.info(f"Set experiment config: {config}")

    def add_round_metrics(
        self,
        round_num: int,
        train_loss: float,
        test_metrics: Optional[Dict[str, float]] = None,
        aggregation_info: Optional[Dict[str, Any]] = None,
        client_metrics: Optional[List[Dict[str, Any]]] = None
    ):
        """
        Add metrics for a training round.

        Args:
            round_num: Round number
            train_loss: Average training loss
            test_metrics: Test set evaluation metrics (NDCG, Hit Rate, etc.)
            aggregation_info: Aggregation statistics
            client_metrics: Per-client metrics
        """
```

```
round_data = {
    "round": round_num,
    "train_loss": train_loss,
    "test_metrics": test_metrics or {},
    "aggregation": aggregation_info or {},
    "client_metrics": client_metrics or [],
    "timestamp": datetime.now().isoformat()
}

self.experiment_data["rounds"].append(round_data)
logger.info(f"Added metrics for round {round_num}")

def add_final_metrics(self, metrics: Dict[str, Any]):
    """
    Add final evaluation metrics.

    Args:
        metrics: Final metrics dictionary
    """
    self.experiment_data["final_metrics"] = metrics
    logger.info(f"Added final metrics: {metrics}")

def add_client_metrics(self, client_id: str, metrics: Dict[str, Any]):
    """
    Add per-client metrics.

    Args:
        client_id: Client identifier
        metrics: Client-specific metrics
    """
    client_data = {
        "client_id": client_id,
        **metrics,
        "timestamp": datetime.now().isoformat()
    }
    self.experiment_data["client_metrics"].append(client_data)

def add_mobile_metrics(self, device_id: str, metrics: Dict[str, Any]):
    """
    Add mobile device resource metrics.

    Args:
        device_id: Device identifier
        metrics: Resource metrics (battery, memory, CPU, etc.)
    """
    mobile_data = {
        "device_id": device_id,
        **metrics,
        "timestamp": datetime.now().isoformat()
    }
    self.experiment_data["mobile_metrics"].append(mobile_data)

def save_json(self, filename: Optional[str] = None) -> str:
    """
    Save metrics to JSON file.

    Args:
        filename: Optional custom filename (default: experiment_id.json)

    Returns:
        Path to saved file
    """
    if filename is None:
        filename = f"{self.experiment_id}.json"

    filepath = self.results_dir / filename

    with open(filepath, 'w') as f:
        json.dump(self.experiment_data, f, indent=2)

    logger.info(f"Saved metrics to {filepath}")
    return str(filepath)

def save_csv_summary(self, filename: Optional[str] = None) -> str:
```

```

"""
Save summary to CSV file (useful for quick analysis).

Args:
    filename: Optional custom filename

Returns:
    Path to saved file
"""
if filename is None:
    filename = f"{self.experiment_id}_summary.csv"

filepath = self.results_dir / filename

# Create summary CSV with key metrics per round
with open(filepath, 'w', newline='') as f:
    writer = csv.writer(f)

    # Header
    writer.writerow([
        'Round', 'Train_Loss', 'NDCG@10', 'Hit@10', 'Precision@10',
        'Recall@10', 'MSE', 'MAE', 'Accuracy', 'Num_Clients', 'Total_Samples'
    ])

    # Round data
    for round_data in self.experiment_data["rounds"]:
        test_metrics = round_data.get("test_metrics", {})
        agg = round_data.get("aggregation", {})

        writer.writerow([
            round_data["round"],
            round_data.get("train_loss", ""),
            test_metrics.get("NDCG@10", ""),
            test_metrics.get("Hit@10", ""),
            test_metrics.get("Precision@10", ""),
            test_metrics.get("Recall@10", ""),
            test_metrics.get("mse", ""),
            test_metrics.get("mae", ""),
            test_metrics.get("accuracy", ""),
            agg.get("num_clients", ""),
            agg.get("total_samples", "")
        ])

    # Final metrics row
    final = self.experiment_data.get("final_metrics", {})
    writer.writerow([
        "FINAL",
        "",
        final.get("NDCG@10", ""),
        final.get("Hit@10", ""),
        final.get("Precision@10", ""),
        final.get("Recall@10", ""),
        final.get("mse", ""),
        final.get("mae", ""),
        final.get("accuracy", ""),
        "",
        ""
    ])

    logger.info(f"Saved CSV summary to {filepath}")
    return str(filepath)

def get_summary(self) -> Dict[str, Any]:
    """
    Get summary statistics of collected metrics.

    Returns:
        Summary dictionary
    """
    rounds = self.experiment_data["rounds"]

    if not rounds:
        return {"message": "No rounds collected yet"}

```

```

    # Extract metrics across rounds
    train_losses = [r["train_loss"] for r in rounds if "train_loss" in r]
    ndcg_scores = [r["test_metrics"].get("NDCG@10", 0) for r in rounds if "test_metrics" in r]
    hit_scores = [r["test_metrics"].get("Hit@10", 0) for r in rounds if "test_metrics" in r]

    return {
        "experiment_id": self.experiment_id,
        "num_rounds": len(rounds),
        "final_train_loss": train_losses[-1] if train_losses else None,
        "final_ndcg@10": ndcg_scores[-1] if ndcg_scores else None,
        "final_hit@10": hit_scores[-1] if hit_scores else None,
        "best_ndcg@10": max(ndcg_scores) if ndcg_scores else None,
        "best_hit@10": max(hit_scores) if hit_scores else None,
        "config": self.experiment_data["config"]
    }

def create_experiment_id(
    dp_epsilon: Optional[float] = None,
    alpha: Optional[float] = None,
    embedding_dim: int = 16,
    num_clients: int = 3,
    seed: int = 42
) -> str:
    """
    Create standardized experiment ID.

    Args:
        dp_epsilon: DP budget (None for no DP)
        alpha: Heterogeneity parameter
        embedding_dim: Model embedding dimension
        num_clients: Number of clients
        seed: Random seed

    Returns:
        Experiment ID string
    """
    parts = []

    if dp_epsilon is None:
        parts.append("dp_inf")
    else:
        parts.append(f"dp_{dp_epsilon}")

    if alpha is not None:
        parts.append(f"alpha_{alpha}")

    parts.append(f"dim_{embedding_dim}")
    parts.append(f"clients_{num_clients}")
    parts.append(f"seed_{seed}")

    return "_".join(parts)

```

File: scripts/rdp_accountant.py

```

"""
RDP (Renyi Differential Privacy) Accountant for DP-SGD.

This module provides privacy accounting for federated learning with DP-SGD.
It computes the privacy budget (epsilon) consumed given the noise multiplier,
clipping norm, number of rounds, and other parameters.
"""

import numpy as np
from typing import Tuple, Optional
import math

class RDPAccountant:
    """
    RDP Accountant for tracking privacy budget in federated learning.

    Implements the RDP composition theorem to compute ( $\epsilon$ ,  $\delta$ )-DP guarantees.
    """

```

```

"""

def __init__(self, delta: float = 1e-5):
    """
    Initialize RDP accountant.

    Args:
        delta: Delta parameter for  $(\epsilon, \delta)$ -DP (default: 1e-5)
    """
    self.delta = delta
    self.orders = np.array([1 + x / 10.0 for x in range(1, 100)] + list(range(12, 65)))

def compute_rdp(self, sigma: float, alpha: float) -> float:
    """
    Compute RDP for Gaussian mechanism.

    Args:
        sigma: Noise multiplier
        alpha: Renyi order (alpha)

    Returns:
        RDP value
    """
    if sigma == 0:
        return float('inf')

    # RDP for Gaussian mechanism:  $\alpha / (2 * \sigma^2)$ 
    return alpha / (2 * sigma ** 2)

def compute_epsilon(self,
                    sigma: float,
                    num_rounds: int,
                    samples_per_round: int,
                    batch_size: int,
                    clip_norm: float = 1.0) -> Tuple[float, float]:
    """
    Compute epsilon (privacy budget) for federated learning with DP-SGD.

    Args:
        sigma: Noise multiplier
        num_rounds: Number of federated learning rounds
        samples_per_round: Average number of samples per client per round
        batch_size: Batch size for local training
        clip_norm: Gradient clipping norm

    Returns:
        Tuple of (epsilon, best_alpha) - epsilon value and the Renyi order that minimizes it
    """
    if sigma == 0 or sigma == float('inf'):
        return (float('inf'), 0.0)

    # Number of steps per round (approximate)
    steps_per_round = max(1, samples_per_round // batch_size)
    total_steps = num_rounds * steps_per_round

    # Compute RDP for each order
    epsilons = []
    for alpha in self.orders:
        # RDP per step
        rdp_per_step = self.compute_rdp(sigma, alpha)

        # Compose over total steps
        total_rdp = total_steps * rdp_per_step

        # Convert RDP to  $(\epsilon, \delta)$ -DP
        epsilon = total_rdp + np.log(1 / self.delta) / (alpha - 1)
        epsilons.append(epsilon)

    # Find minimum epsilon (best privacy guarantee)
    min_epsilon = min(epsilons)
    best_alpha_idx = np.argmin(epsilons)
    best_alpha = self.orders[best_alpha_idx]

    return (min_epsilon, best_alpha)

```

```

def compute_sigma_for_epsilon(self,
                              target_epsilon: float,
                              num_rounds: int,
                              samples_per_round: int,
                              batch_size: int,
                              clip_norm: float = 1.0,
                              sigma_min: float = 0.1,
                              sigma_max: float = 200.0,
                              tolerance: float = 0.1) -> Optional[float]:
    """
    Find noise multiplier (sigma) that achieves target epsilon.

    Uses binary search to find the sigma value that gives the desired epsilon.

    Args:
        target_epsilon: Target privacy budget
        num_rounds: Number of federated learning rounds
        samples_per_round: Average number of samples per client per round
        batch_size: Batch size for local training
        clip_norm: Gradient clipping norm
        sigma_min: Minimum sigma to search
        sigma_max: Maximum sigma to search
        tolerance: Tolerance for epsilon match

    Returns:
        Sigma value that achieves target epsilon, or None if not found
    """
    if target_epsilon == float('inf'):
        return 0.0

    # Binary search for sigma
    low, high = sigma_min, sigma_max

    for _ in range(50): # Max 50 iterations
        mid = (low + high) / 2
        epsilon, _ = self.compute_epsilon(mid, num_rounds, samples_per_round, batch_size, clip_norm)

        if abs(epsilon - target_epsilon) < tolerance:
            return mid

        if epsilon > target_epsilon:
            # Need more noise (higher sigma)
            low = mid
        else:
            # Can use less noise (lower sigma)
            high = mid

    # Return the best approximation
    final_sigma = (low + high) / 2
    final_epsilon, _ = self.compute_epsilon(final_sigma, num_rounds, samples_per_round, batch_size, clip_norm)

    if abs(final_epsilon - target_epsilon) < tolerance * 2:
        return final_sigma

    return None

def compute_sigma_for_target_epsilon(target_epsilon: float,
                                     num_rounds: int = 10,
                                     samples_per_client: int = 100,
                                     batch_size: int = 32,
                                     clip_norm: float = 1.0,
                                     delta: float = 1e-5) -> Optional[float]:
    """
    Convenience function to compute sigma for target epsilon.

    Args:
        target_epsilon: Target privacy budget
        num_rounds: Number of federated learning rounds
        samples_per_client: Average samples per client
        batch_size: Batch size
        clip_norm: Gradient clipping norm

```

```

    delta: Delta for ( $\epsilon$ ,  $\delta$ )-DP

Returns:
    Recommended sigma value
"""
accountant = RDPAccountant(delta=delta)
return accountant.compute_sigma_for_epsilon(
    target_epsilon=target_epsilon,
    num_rounds=num_rounds,
    samples_per_round=samples_per_client,
    batch_size=batch_size,
    clip_norm=clip_norm
)

def compute_epsilon_for_sigma(sigma: float,
                              num_rounds: int = 10,
                              samples_per_client: int = 100,
                              batch_size: int = 32,
                              clip_norm: float = 1.0,
                              delta: float = 1e-5) -> Tuple[float, float]:
    """
    Convenience function to compute epsilon for given sigma.

    Args:
        sigma: Noise multiplier
        num_rounds: Number of federated learning rounds
        samples_per_client: Average samples per client
        batch_size: Batch size
        clip_norm: Gradient clipping norm
        delta: Delta for ( $\epsilon$ ,  $\delta$ )-DP

    Returns:
        Tuple of (epsilon, best_alpha)
    """
    accountant = RDPAccountant(delta=delta)
    return accountant.compute_epsilon(
        sigma=sigma,
        num_rounds=num_rounds,
        samples_per_round=samples_per_client,
        batch_size=batch_size,
        clip_norm=clip_norm
    )

```

File: scripts/recommendation_metrics.py

```

"""
Recommendation Metrics for Movie Recommendation Evaluation
Implements NDCG@10, Hit@10, Precision, Recall as required by the thesis expose.
"""

import numpy as np
import torch
from typing import List, Tuple, Dict

def ndcg_at_k(relevance_scores: List[float], k: int = 10) -> float:
    """
    Compute Normalized Discounted Cumulative Gain at rank k.

    Args:
        relevance_scores: List of relevance scores (ratings) for top-k items
        k: Rank cutoff (default 10)

    Returns:
        NDCG@k score (0.0 to 1.0)
    """
    if len(relevance_scores) == 0 or k == 0:
        return 0.0

    # Take top k
    scores = relevance_scores[:k]

```

```
# Compute DCG
dcg = sum(score / np.log2(idx + 2) for idx, score in enumerate(scores))

# Compute ideal DCG (perfect ranking)
ideal_scores = sorted(scores, reverse=True)
idcg = sum(score / np.log2(idx + 2) for idx, score in enumerate(ideal_scores))

if idcg == 0:
    return 0.0

return dcg / idcg

def hit_rate_at_k(relevant_items: List[bool], k: int = 10) -> float:
    """
    Compute Hit Rate at rank k.

    Args:
        relevant_items: List of boolean values indicating relevance in top-k
        k: Rank cutoff (default 10)

    Returns:
        Hit@k rate (0.0 to 1.0)
    """
    if len(relevant_items) == 0 or k == 0:
        return 0.0

    # Take top k
    top_k_relevant = relevant_items[:k]

    # Hit rate = 1 if any item in top-k is relevant, else 0
    return 1.0 if any(top_k_relevant) else 0.0

def precision_at_k(relevant_items: List[bool], k: int = 10) -> float:
    """
    Compute Precision at rank k.

    Args:
        relevant_items: List of boolean values indicating relevance in top-k
        k: Rank cutoff (default 10)

    Returns:
        Precision@k (0.0 to 1.0)
    """
    if len(relevant_items) == 0 or k == 0:
        return 0.0

    # Take top k
    top_k_relevant = relevant_items[:k]

    # Precision = relevant items / total items in top-k
    return sum(top_k_relevant) / len(top_k_relevant)

def recall_at_k(relevant_items: List[bool], total_relevant: int, k: int = 10) -> float:
    """
    Compute Recall at rank k.

    Args:
        relevant_items: List of boolean values indicating relevance in top-k
        total_relevant: Total number of relevant items for this user
        k: Rank cutoff (default 10)

    Returns:
        Recall@k (0.0 to 1.0)
    """
    if total_relevant == 0 or k == 0:
        return 0.0

    # Take top k
    top_k_relevant = relevant_items[:k]

    # Recall = relevant items found / total relevant items
```



```

return sum(top_k_relevant) / total_relevant

def evaluate_recommendations(
    model: torch.nn.Module,
    test_data: List[Tuple[int, int, float]],
    num_users: int,
    num_items: int,
    k: int = 10,
    all_items: List[int] = None
) -> Dict[str, float]:
    """
    Evaluate recommendation model using standard metrics.

    Args:
        model: PyTorch model with predict(user_id, item_id) method
        test_data: List of (user_id, item_id, rating) tuples for testing
        num_users: Total number of users
        num_items: Total number of items
        k: Rank cutoff for metrics (default 10)
        all_items: List of all item IDs to consider for ranking (if None, uses all items)

    Returns:
        Dictionary with evaluation metrics: NDCG@k, Hit@k, Precision@k, Recall@k
    """
    model.eval()

    # Group test data by user
    user_test_items = {}
    user_ratings = {}

    for user_id, item_id, rating in test_data:
        if user_id not in user_test_items:
            user_test_items[user_id] = []
            user_ratings[user_id] = []
        user_test_items[user_id].append(item_id)
        user_ratings[user_id].append(rating)

    # Get all items if not provided
    if all_items is None:
        all_items = list(range(num_items))

    ndcg_scores = []
    hit_scores = []
    precision_scores = []
    recall_scores = []

    with torch.no_grad():
        for user_id in user_test_items.keys():
            # Get ground truth relevant items for this user
            ground_truth_items = set(user_test_items[user_id])
            ground_truth_ratings = dict(zip(user_test_items[user_id], user_ratings[user_id]))

            # Score all items for this user
            item_scores = []
            for item_id in all_items:
                # Get prediction
                user_tensor = torch.LongTensor([user_id])
                item_tensor = torch.LongTensor([item_id])

                if hasattr(model, 'predict'):
                    score = model.predict(user_tensor, item_tensor).item()
                else:
                    score = model(user_tensor, item_tensor).item()

                item_scores.append((item_id, score))

            # Sort by score (descending)
            item_scores.sort(key=lambda x: x[1], reverse=True)

            # Get top-k predictions
            top_k_items = [item_id for item_id, _ in item_scores[:k]]
            top_k_scores = [score for _, score in item_scores[:k]]

```

```

    # Check relevance of top-k (relevant = rating >= 4.0)
    top_k_relevant = [
        item_id in ground_truth_items and ground_truth_ratings.get(item_id, 0.0) >= 4.0
        for item_id in top_k_items
    ]

    # Get relevance scores for NDCG (use actual ratings 1-5 as relevance)
    relevance_scores = []
    for item_id in top_k_items:
        if item_id in ground_truth_items:
            # Use actual rating (1-5) as relevance score for NDCG
            rating = ground_truth_ratings.get(item_id, 0.0)
            relevance = rating
        else:
            relevance = 0.0
        relevance_scores.append(relevance)

    # Compute metrics for this user
    ndcg_scores.append(ndcg_at_k(relevance_scores, k))
    hit_scores.append(hit_rate_at_k(top_k_relevant, k))
    precision_scores.append(precision_at_k(top_k_relevant, k))
    recall_scores.append(recall_at_k(top_k_relevant, len(ground_truth_items), k))

# Average over all users
return {
    f'NDCG@{k}': np.mean(ndcg_scores) if ndcg_scores else 0.0,
    f'Hit@{k}': np.mean(hit_scores) if hit_scores else 0.0,
    f'Precision@{k}': np.mean(precision_scores) if precision_scores else 0.0,
    f'Recall@{k}': np.mean(recall_scores) if recall_scores else 0.0,
    'num_users_evaluated': len(ndcg_scores)
}

def evaluate_recommendations_simple(
    model: torch.nn.Module,
    test_data: List[Tuple[int, int, float]],
    num_users: int,
    num_items: int,
    k: int = 10
) -> Dict[str, float]:
    """
    Simplified evaluation for faster computation (samples users instead of all).

    Args:
        model: PyTorch model
        test_data: List of (user_id, item_id, rating) tuples
        num_users: Total number of users
        num_items: Total number of items
        k: Rank cutoff

    Returns:
        Dictionary with evaluation metrics
    """
    # Sample users for evaluation (to speed up)
    unique_users = list(set(user_id for user_id, _, _ in test_data))

    # Evaluate on sampled users
    sampled_data = []
    for user_id, item_id, rating in test_data:
        if user_id in unique_users[:min(100, len(unique_users))]: # Sample up to 100 users
            sampled_data.append((user_id, item_id, rating))

    return evaluate_recommendations(model, sampled_data, num_users, num_items, k)

```